Inorganic Cumulative Exam

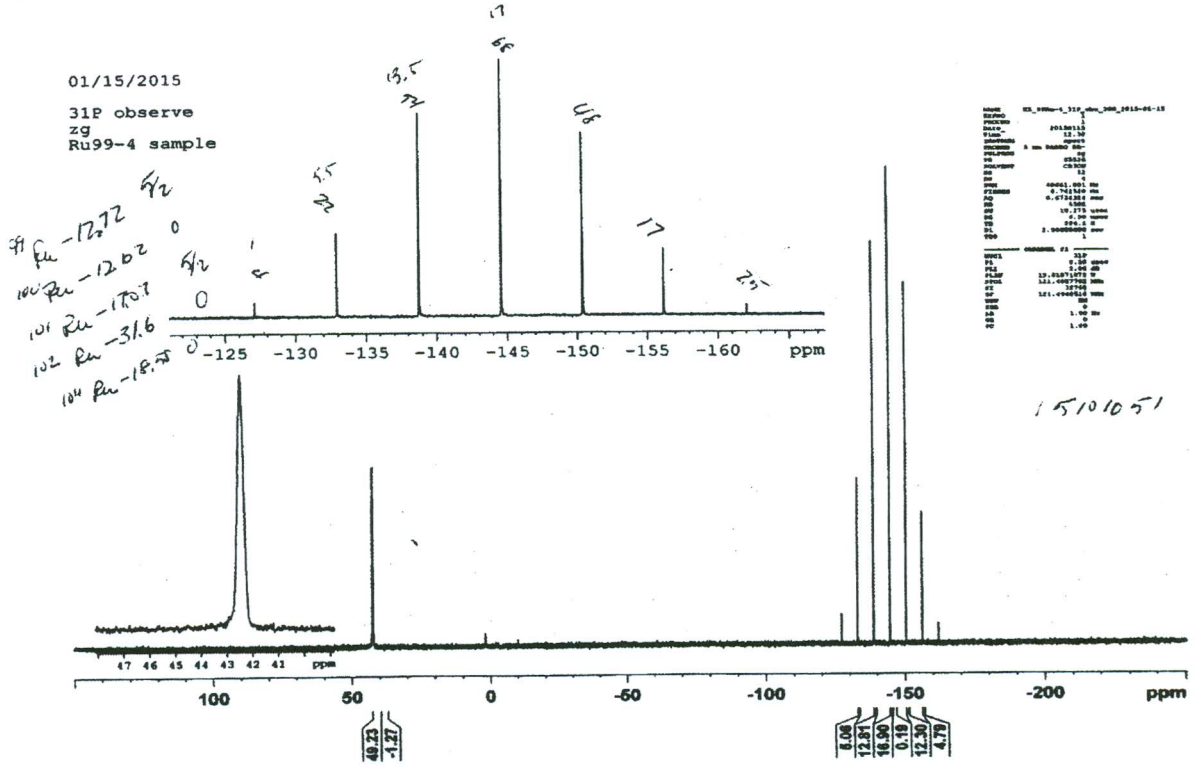
February 13, 2015

1. The following reactions were performed with the goal of producing a homonuclear dimer.

   a. $Ru(CO)_4Cl_2$ + LL = $Cl_2(CO)_2Ru\text{-}LL\text{-}Ru(CO)_2Cl_2$

   b. $Cl_2(CO)_2Ru\text{-}LL\text{-}Ru(CO)_2Cl_2$ = $Cl(CO)(L)Ru\text{-}LL\text{-}Ru(CO)LCl$

   c. $Cl(CO)(L)Ru\text{-}LL\text{-}Ru(CO)LCl$ = $L_2Ru\text{-}LL\text{-}RuL_2$

Where LL is a symmetric bridging ligand acting as a bidentate ligand to each metal and L is a simple bidentate ligand. The reaction was readily followed by IR spectroscopy. What spectroscopic features and associated changes in the reactants and products would you expect to see that are diagnostic for the indicated reactions?

2. The ^{31}P NMR of $Ru(bpy)_2PPh_3Cl^+$ is shown below. Given that ^{31}P has a spin of ½ and ^{99}Ru has a spin of 5/2 with a natural abundance of 13% and ^{101}Ru has a spin of 5/2 with a natural abundance of 17% rationalize the multiplet at -150 ppm. PPh3 is triphenylphosphine and bpy is bipyridine.